CO2035

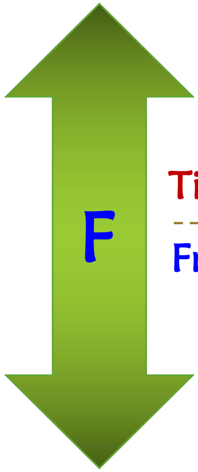
5. Discrete Fourier Transform (DFT)

Contents

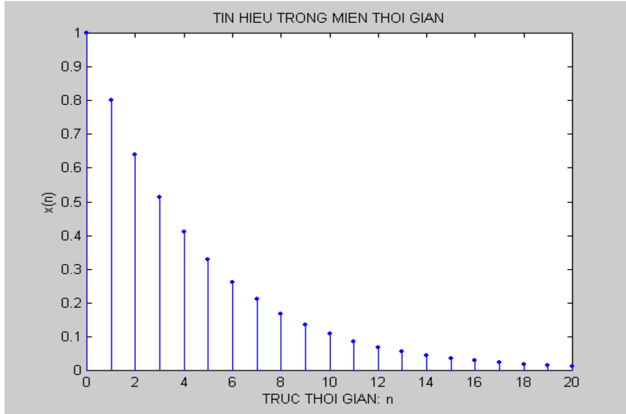
Introduction to DFT

- Continuous Fourier Transform

x(n)

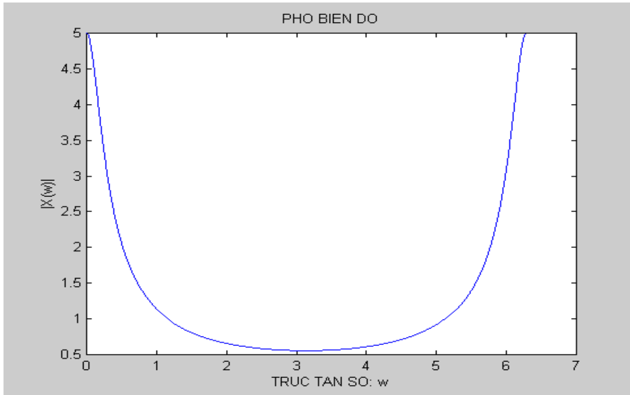


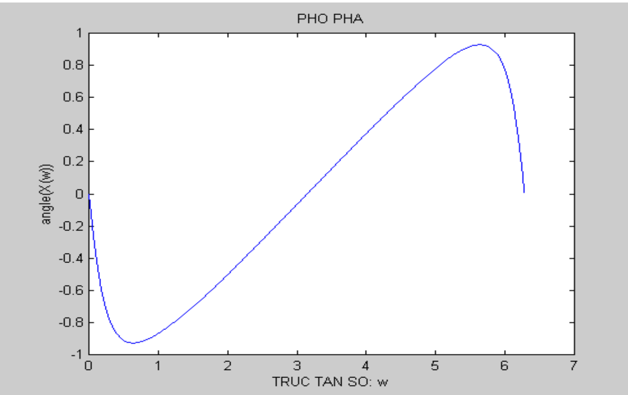
$x(n) = 0.8^n u(n)$



Time-Domain

Frequency Domain





$$X(\omega) = \sum_{n=-\infty}^{+\infty} x(n)e^{-j\omega n}$$

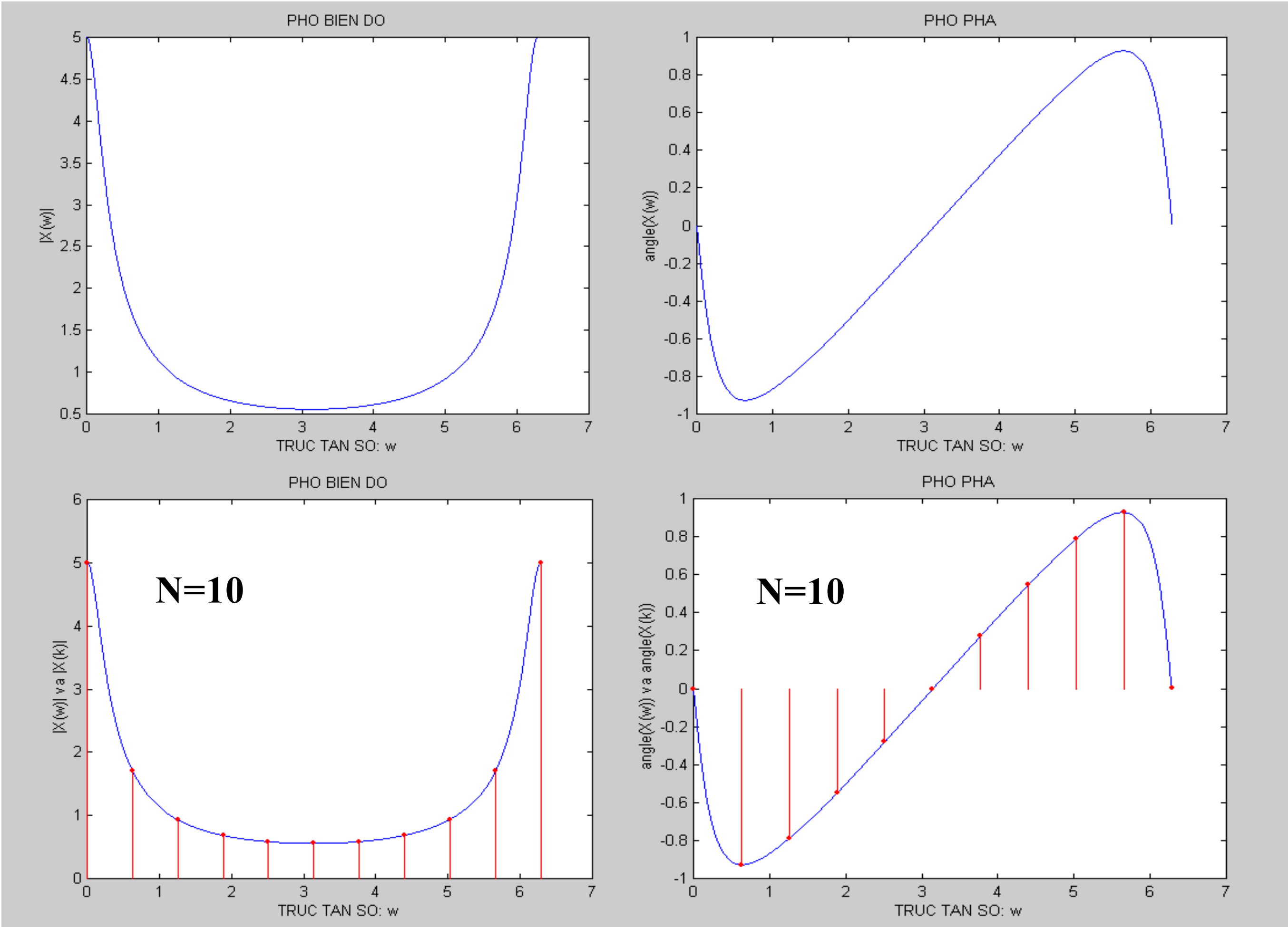
- Problem: X(ω) is continuous by frequency ω → it is not appropriate for processing in computers

Frequency Domain Sampling

$$\mathbf{X(\omega)}$$

Sampling

$$\mathbf{X(k) \equiv X(\omega = \frac{2\pi}{N}k)}$$



Frequency Domain Sampling

$$X(\omega)\Big|_{\omega=2\pi\frac{k}{N}} = X\left(2\pi\frac{k}{N}\right) = \sum_{n=-\infty}^{+\infty} x(n)e^{-j2\pi\frac{k}{N}n} \qquad k = 0, 1, \dots N-1$$

$$X(k) = \dots + \sum_{n=-N}^{-1} x(n)e^{-j2\pi\frac{k}{N}n} + \sum_{n=0}^{N-1} x(n)e^{-j2\pi\frac{k}{N}n} + \sum_{n=N}^{2N-1} x(n)e^{-j2\pi\frac{k}{N}n} + \dots = \sum_{i=-\infty}^{+\infty}\sum_{n=iN}^{iN+N-1} x(n)e^{-j2\pi\frac{k}{N}n}$$

Replace n by (n – iN)

$$X(k) = \sum_{i=-\infty}^{+\infty}\sum_{n=0}^{N-1} x(n-iN)e^{-j2\pi\frac{k}{N}n} = \sum_{n=0}^{N-1}\left[\sum_{i=-\infty}^{+\infty} x(n-iN)\right]e^{-j2\pi\frac{k}{N}n}$$

Let $x_p(n) = \sum_{i=-\infty}^{+\infty} x(n-iN)$

$$\Rightarrow X(k) = \sum_{n=0}^{N-1} x_p(n)e^{-j2\pi\frac{k}{N}n}$$

$$x_p(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi\frac{k}{N}n} \qquad n = 0, 1, \dots, N-1$$

$$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x_p(n)e^{-j2\pi\frac{k}{N}n} \qquad k = 0, 1, \dots, N-1$$

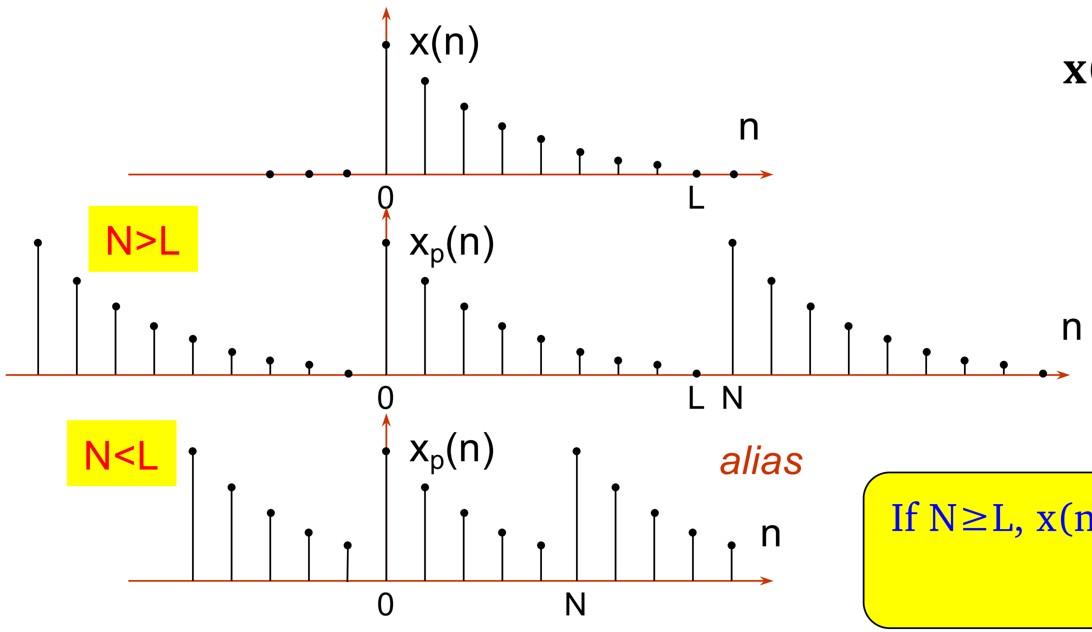
- $x_p(n)$: the periodic repetition of x(n) every N samples (it is clearly periodic with fundamental period N)

Frequency Domain Sampling

$$c_k = \frac{1}{N} X(k) \quad k = 0, 1, \dots, N-1$$

$$x_p(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi \frac{k}{N} n} \quad n = 0, 1, \dots, N-1$$

It is able to recover signal $x_p(n)$ from the samples of $X(\omega)$



$$x(n) = \begin{cases} x_p(n) & 0 \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

If N≥L, x(n) can be recovered from its samples at frequencies $\omega_k = \frac{2\pi k}{N}$

Frequency Domain Sampling

- $X(\omega)$ can be recovered from its samples $X(k)$ for $k = 0, 1, \ldots, N\text{-}1$
  - Asume that $N \geq L$, then $x(n) = x_p(n)$ for $0 \leq n \leq N\text{-}1$

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j2\pi\frac{k}{N}n}$$

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x(n)e^{-j\omega n} = \sum_{n=0}^{N-1}\left[\frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j2\pi\frac{k}{N}n}\right]e^{-j\omega n} = \sum_{k=0}^{N-1} X(k)\left[\frac{1}{N}\sum_{n=0}^{N-1} e^{-j(\omega-2\pi\frac{k}{N})n}\right]$$

- Let

$$P(\omega) = \frac{1}{N}\sum_{n=0}^{N-1} e^{-j\omega n} = \frac{1}{N}\frac{1-e^{-j\omega N}}{1-e^{-j\omega}} = \frac{\sin\left(\frac{\omega N}{2}\right)}{N\sin\left(\frac{\omega}{2}\right)}e^{-j\omega\frac{(N-1)}{2}}$$

$$\Longrightarrow P\left(\frac{2\pi}{N}k\right) = \begin{cases} 1 & k = 0 \\ 0 & k = 1,2,\ldots,N-1 \end{cases}$$

- Then,

$$X(\omega) = \sum_{k=0}^{N-1} X(k)P\left(\omega - \frac{2\pi}{N}k\right) \qquad N \geq L$$

Frequency Domain Sampling

x(n)
Length L≤N

F

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x(n)e^{-j\omega n}$$

Sampling

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi\frac{k}{N}n}$$

Reconstruction

$$X(\omega) = \sum_{k=0}^{N-1} X(k)P(\omega - \omega_k)$$

where $\begin{cases} P(\omega) = \dfrac{1}{N}\displaystyle\sum_{k=0}^{N-1} e^{-j\omega n} \\ \omega_k = \dfrac{2\pi k}{N} \end{cases}$

Reconstruction

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j2\pi\frac{k}{N}n}$$

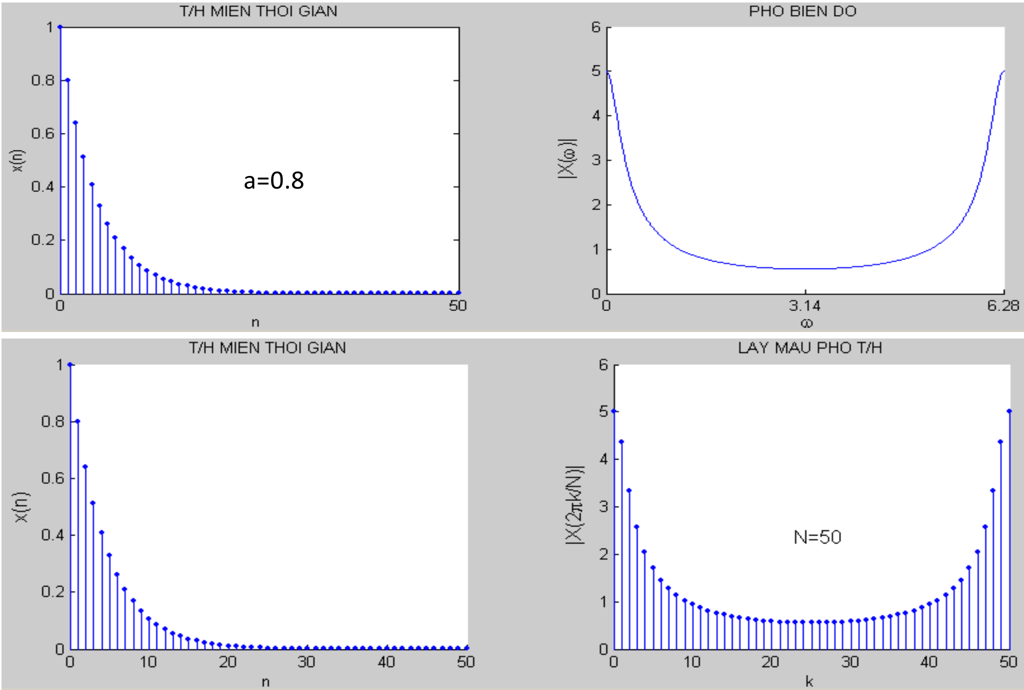
Frequency Domain Sampling

- Example: $x(n) = a^n u(n)$, $0<a<1$
  - The spectrum of this signal is sampled at frequencies $\omega_k = \frac{2\pi k}{N}$ for k=0, 1, …, N-1

$$X(\omega) = \sum_{n=0}^{+\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

$$\Rightarrow X(k) = X\left(\omega = \frac{2\pi k}{N}\right) = \frac{1}{1 - ae^{-j2\pi\frac{k}{N}}}$$

$$x_p(n) = \sum_{i=-\infty}^{+\infty} x(n - iN) = \sum_{i=-\infty}^{0} a^{n-iN} = \frac{a^n}{1 - a^N}$$



Discrete Fourier Transform (DFT)

DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi\frac{k}{N}n} \qquad k = 0, 1, \dots, N-1$$

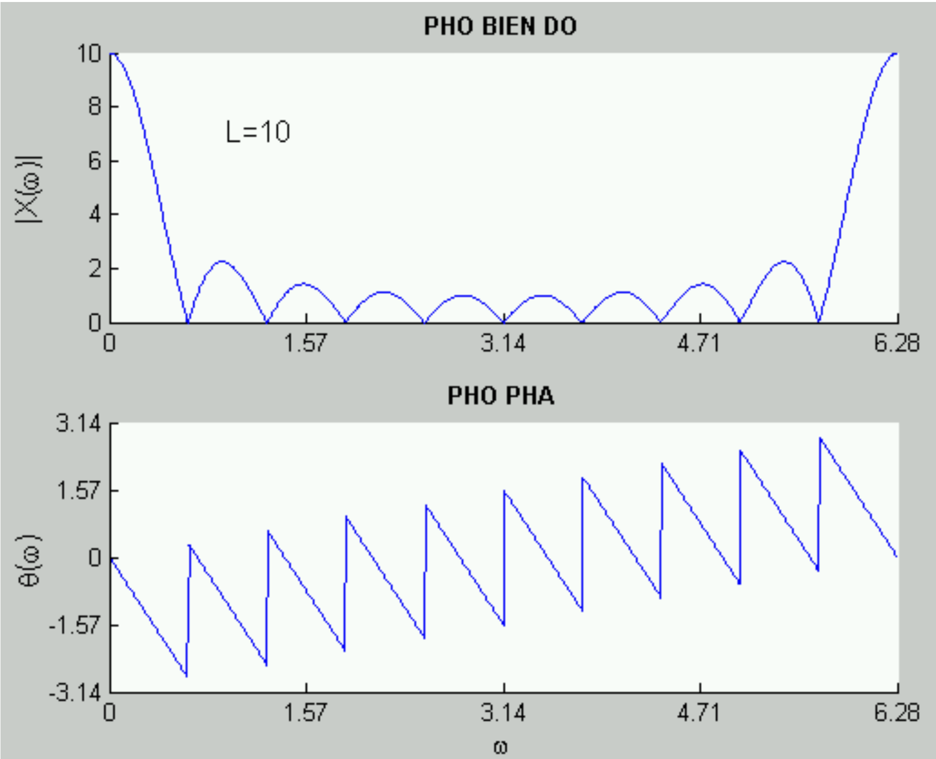
IDFT

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi\frac{k}{N}n} \qquad n = 0, 1, \dots, N-1$$

Discrete Fourier Transform

- Examples: determine N-point DFT of the sequence x(n) of length L (N≥L)

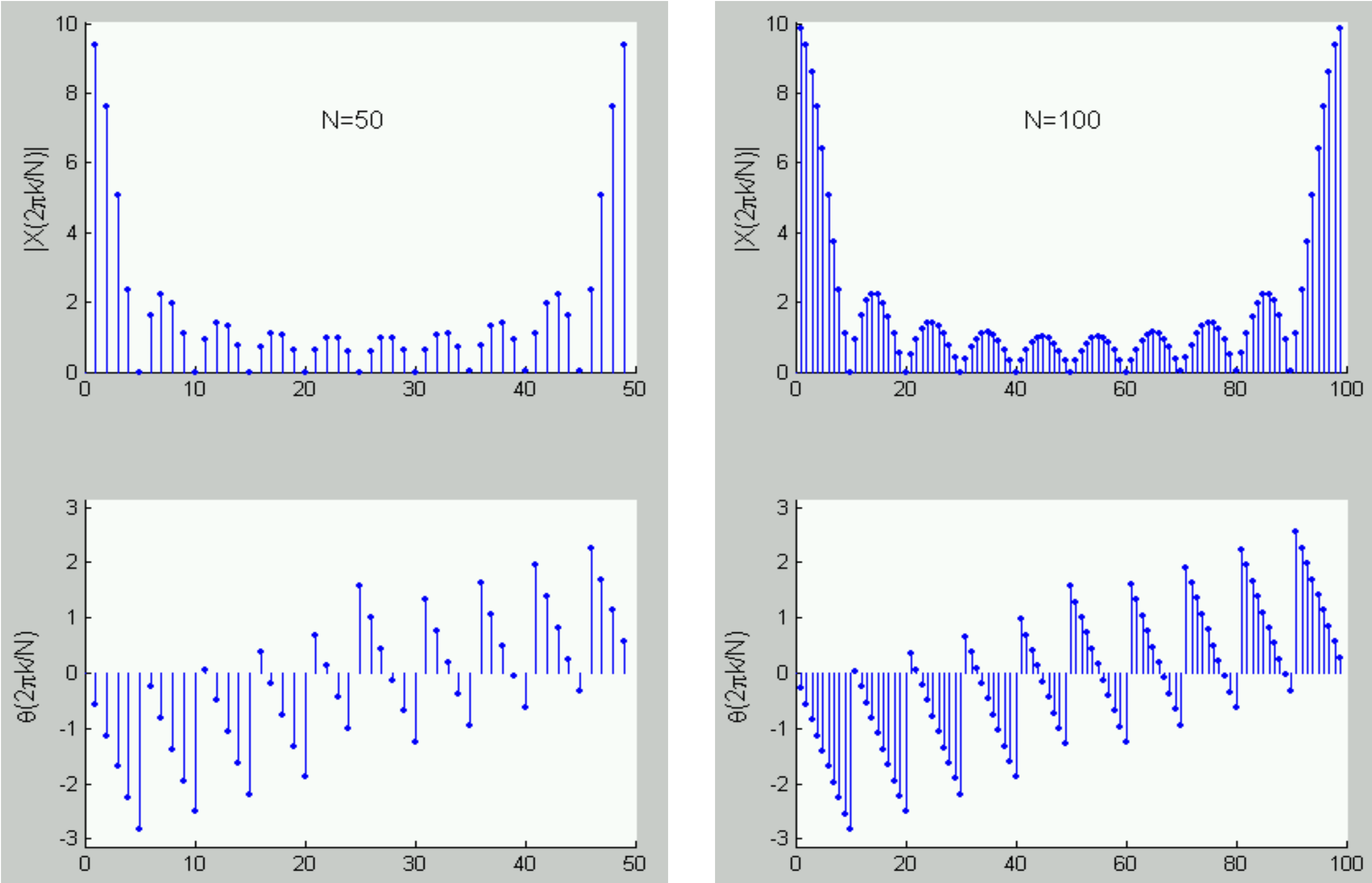
$$x(n) = \begin{cases} 1 & 0 \le n \le L-1 \\ 0 & \text{otherwise} \end{cases}$$



$$\mathbf{X(\omega) = \sum_{n=-\infty}^{+\infty} x(n)e^{-j\omega n} = \sum_{n=0}^{L-1} e^{-j\omega n}}$$

$$\mathbf{X(\omega) = \frac{1 - e^{-j\omega L}}{1 - e^{-j\omega}} = \frac{\sin\left(\frac{\omega L}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} e^{-j\omega\frac{(L-1)}{2}}}$$

Discrete Fourier Transform



DFT – Linear Transform

DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{k}{N} n}$$

$$k = 0, 1, \dots, N-1$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$k = 0, 1, \dots, N-1$$

$$W_N = e^{-j\frac{2\pi}{N}}$$

IDFT

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi \frac{k}{N} n}$$

$$n = 0, 1, \dots, N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

$$n = 0, 1, \dots, N-1$$

1-Point DFT
- N x complex multiplications
- (N-1) x complex additions

N-point DFT
- N^2 x complex multiplications
- N(N-1) x complex additions

DFT – Linear Transform

Signal sequence in time-domain
$$\mathbf{x}_N = \begin{bmatrix} x(0) \\ x(1) \\ . \\ . \\ . \\ x(N-1) \end{bmatrix}$$

Samples in frequency-domain
$$\mathbf{X}_N = \begin{bmatrix} X(0) \\ X(1) \\ . \\ . \\ . \\ X(N-1) \end{bmatrix}$$

Matrix for linear transform
$$\mathbf{W}_N = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N & W_N^2 & \cdots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

$$W_N^{-1} = \frac{1}{N} W_N^*$$
$$W_N W_N^* = N I_N$$

- N-point DFT

$$\mathbf{X}_N = \mathbf{W}_N \mathbf{x}_N$$

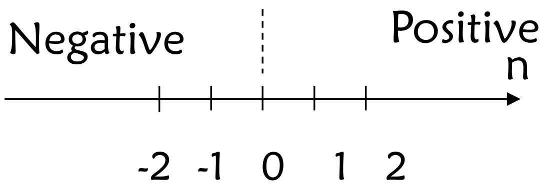
$$\mathbf{x}_N = \mathbf{W}_N^{-1} \mathbf{X}_N$$
$$\mathbf{x}_N = \frac{1}{N} \mathbf{W}_N^* \mathbf{X}_N$$

$$\mathbf{W}_N^{-1} = \frac{1}{N} \mathbf{W}_N^*$$
$$\mathbf{W}_N \mathbf{W}_N^* = N \mathbf{I}_N$$

$\mathbf{W}_N$ is diagonal matrix

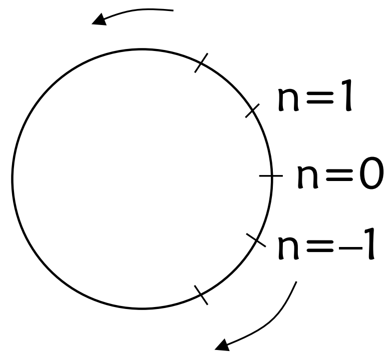
DFT - Circular Representation

Linear



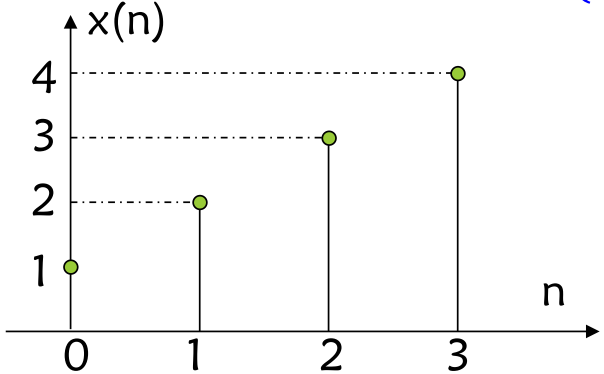
Circular

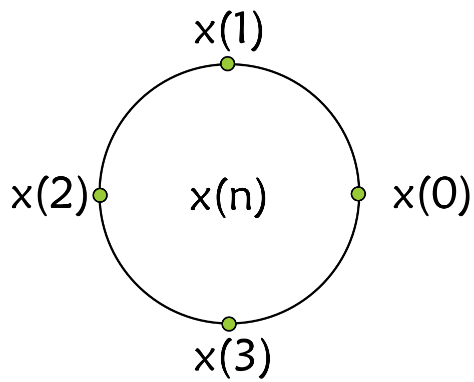
Positive (anticlockwise direction)



Negative (clockwise direction)

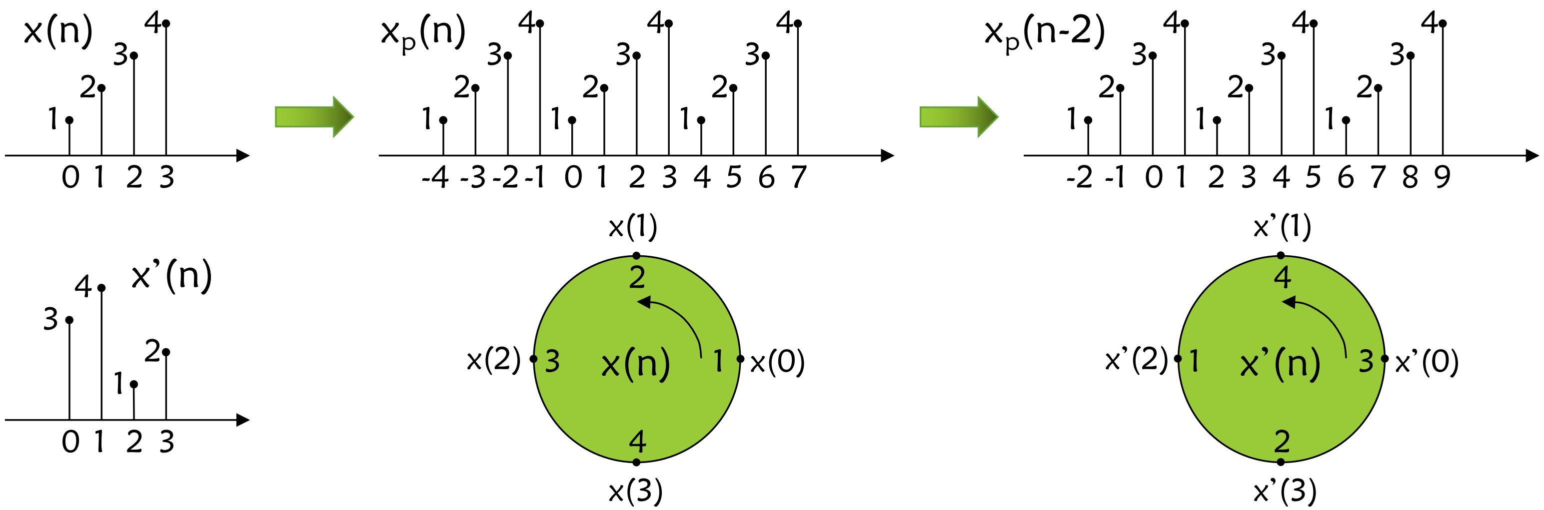
$$\mathbf{x(n) = \{\overset{\uparrow}{1} \quad 2 \quad 3 \quad 4\}}$$





DFT – Circular Representation

- Periodic sequence with period N $\mathbf{x_p(n) = \sum_{i=-\infty}^{+\infty} x(n - iN)}$
- Shifted sequence $\mathbf{x_p^k(n)}$ by k samples $\mathbf{x_p^k(n) = x_p(n-k) = \sum_{i=-\infty}^{+\infty} x(n - iN - k)}$
- Finite length sequence from the shifted sequence $\mathbf{x_p^k(n)}$ $\mathbf{x'(n) = \begin{cases} x_p^k(n) & 0 \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}}$
- Relationship of x(n) and x'(n): **circular shifting** $\mathbf{x'(n) = x(n-k, MOD\ N) \equiv x((n-k))_N}$



DFT – Circular Symmetric

- N-point sequence is called **circularly even** if it is symmetric about the point zero on the circle.
  - i.e. $x(N - n) = x(n),\ 0 \le n \le N - 1$

- N-point sequence is called **circularly odd** if it is antisymmetric about the point zero on the circle
  - i.e. $x(N - n) = -x(n),\ 0 \le n \le N - 1$

- **Time reversal** of N-point sequence is attained by reversing its samples about the point zero on the circle.
  - i.e. $x((-n))_N = x(N - n),\ 0 \le n \le N - 1$
  - It is equivalent to plotting x(n) in a **clockwise direction** on the circle

DFT – Properties

- Periodicity

$$\mathbf{x(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X(k)} \Longrightarrow \begin{cases} \mathbf{x(n) = x(n+N)} & \forall \mathbf{n} \\ \mathbf{X(k) = X(k+N)} & \forall \mathbf{k} \end{cases}$$

- Linearity

$$\begin{cases} \mathbf{x_1(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X_1(k)} \\ \mathbf{x_2(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X_2(k)} \end{cases} \Longrightarrow \mathbf{a_1x_1(n) + a_2x_2(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{a_1X_1(k) + a_2X_2(k)}$$

- Circular Convolution

$$\begin{cases} \mathbf{x_1(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X_1(k)} \\ \mathbf{x_2(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X_2(k)} \end{cases} \Longrightarrow \mathbf{x_1(n)} \; Ⓝ \; \mathbf{x_2(n)} \xleftrightarrow{\mathbf{DFT_N}} \mathbf{X_1(k)X_2(k)}$$

Ⓝ
N-point
circular convolution

$$\mathbf{x_1(n)} \, Ⓝ \, \mathbf{x_2(n)} = \sum_{\mathbf{k=0}}^{\mathbf{N-1}} \mathbf{x_1(k)x_2(n-k)_N} \quad \mathbf{n = 0, 1, \ldots, N-1}$$

DFT – Additional Properties

- Time Reversal

$$x(n) \xleftrightarrow{DFT_N} X(k) \Rightarrow x\big((-n)\big)_N = x(N-n) \xleftrightarrow{DFT_N} X\big((-k)\big)_N = X(N-k)$$

- Circular Time Shift

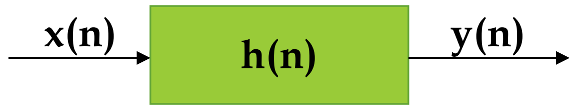
$$x(n) \xleftrightarrow{DFT_N} X(k) \Rightarrow x\big((n-i)\big)_N \xleftrightarrow{DFT_N} X(k)e^{-j2\pi\frac{k}{N}i}$$

- Circular Frequency Shift

$$x(n) \xleftrightarrow{DFT_N} X(k) \Rightarrow x(n)e^{j2\pi\frac{n}{N}i} \xleftrightarrow{DFT_N} X((k-i))_N$$

- Complex Conjugate, Circular Correlation, Multiplication, Parseval's Theorem
  - Textbook, pp. 478-479

DFT – Linear Filtering



$$y(n) = \sum_{k=0}^{M-1} h(k)x(n-k)$$

- $Y(\omega) = H(\omega)X(\omega)$
  - continuous by frequency ω
  - it is not appropriate for processing in computers

  → DFT: a computationally efficient methods of time-domain convolution

- Linear Filtering (for Short Sequence)

  $x(n)$: Length of L ($n=0,1,\ldots,L-1$)
  $h(n)$: Length of M ($n=0,1,\ldots,M-1$)

  ⇒ $y(n)$: Length of $N = M + L - 1$

  $Y(k) = H(k)X(k)$, $k=0,1,\ldots,N-1$

  $H(k)$, $X(k)$: DFT N-point of $h(n)$ and $x(n)$

  (Several zeros padding to $h(n)$ and $x(n)$ be in Length of N if needed)

$y(n) = IDFT_N\{Y(k)\}$

- Circular convolution N-point of $h(n)$ và $x(n)$ ⟺ Linear convolution of $h(n)$ and $x(n)$
- DFT is used to compute linear filtering (several zeros padding to $h(n)$ and $x(n)$)

DFT – Linear Filtering

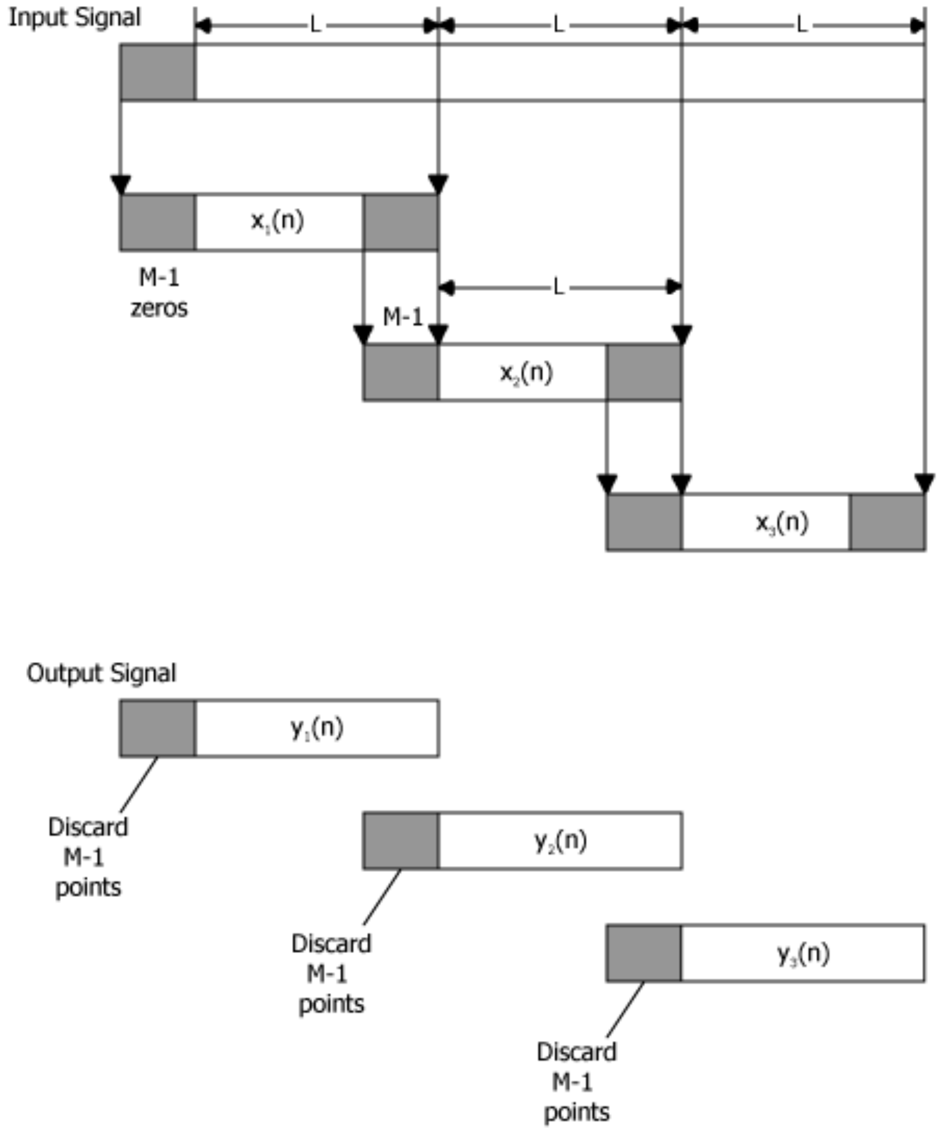
- Summary

$$x(n) \xrightarrow{DFT_N} X(k)$$
$$h(n) \xrightarrow{DFT_N} H(k)$$
$$\times \rightarrow Y(k) = X(k)H(k) \xrightarrow{IDFT_N} y(n)$$

- For long sequence, it will be segmented into fixed-size blocks priors to processing
  - Overlap-Save
  - Overlap-Add
- Assumption
  - Filter h(n) has Length of M
  - Input sequence x(n) are segmented into fixed-size blocks of length L (L >>M)

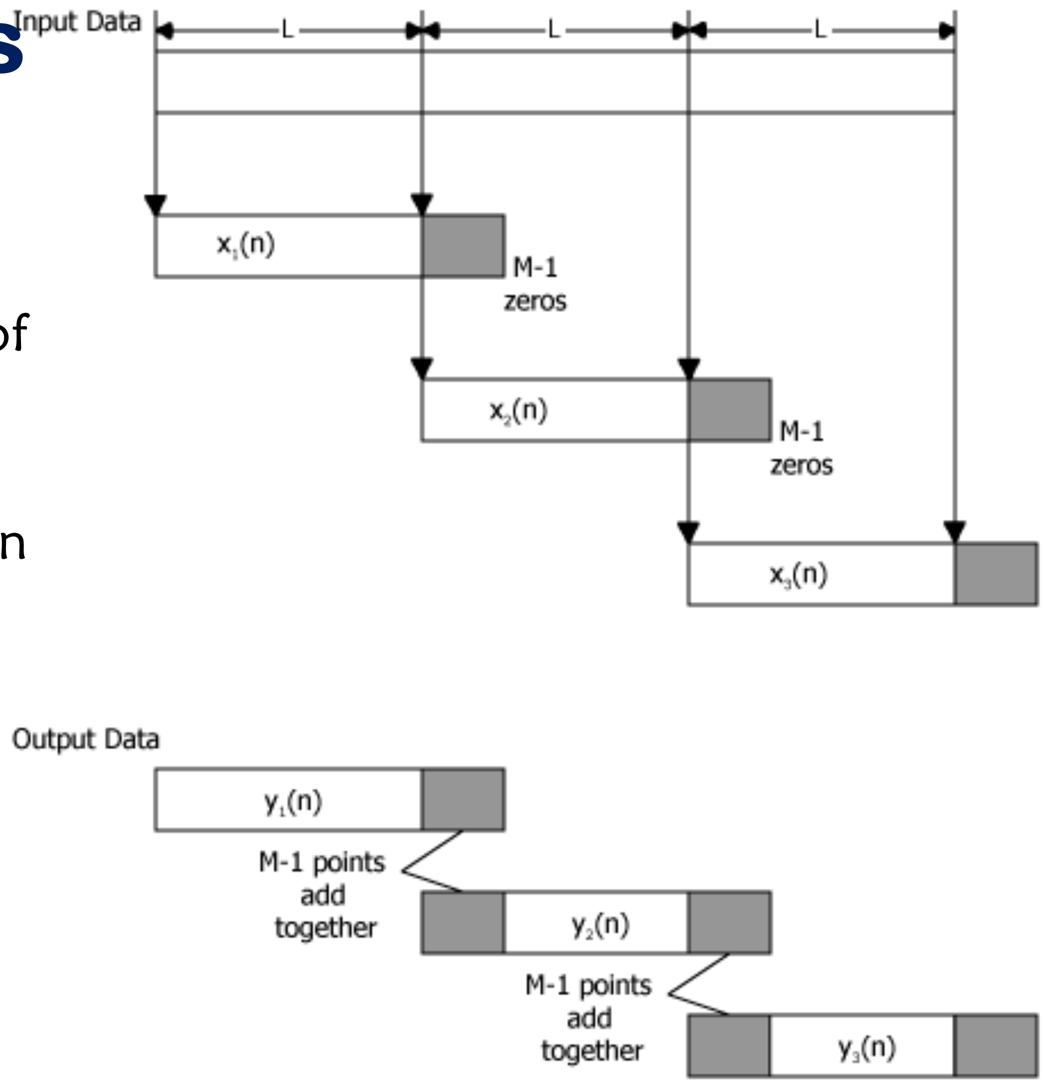
Linear Filtering Methods (1

- Overlap-Save method
  - DFT_N and $IDFT_N$ where $N = L+M-1$
  - Each data block consists of last $(M - 1)$ data points of previous block and L new data points of next block.
  - $(M - 1)$ data points of the first block are zeros
  - $(L - 1)$ zeros padding to $h(n)$ to be in Length of N for calculating DFT N-point.



Linear Filtering Methods

- Overlap-Add method
  - (M – 1) zeros are padded to the end of each input block.
  - (L – 1) zeros padding to h(n) to be in Length of N for calculating DFT N-point.



Exercise